

Gun Violence Cross-Geography Study — Research Design

Working title: Cross-Geography Decoupling of Race and Structural Inequity Effects on Gun Violence Rates in Working-Class American Communities

Status: Pre-registration draft. Not yet executed. All hypotheses, cell definitions, matching criteria, and analytic decisions specified BEFORE data analysis begins.

Pipeline position: Standalone methodology contribution. Substrate for Paper 6 (methodology corpus) extension. Independent of Collatz publication queue timeline.

§0. Pre-registration commitments

This document is the pre-registration. Changes to anything below after data analysis begins must be (a) explicitly noted as post-hoc, (b) justified, (c) reported alongside the original specification.

The design specifies:

- Cell definitions and population thresholds
- Working-class American matching profile
- Matching algorithm
- Sample size targets per cell
- Pre-registered hypotheses with falsification criteria
- Power-stratified claim framework
- Analysis sequence

No looking at gun violence outcome data until all of the above is locked.

§1. Research question

Primary question: When race and structural inequity are decoupled (sampled across

cells where they vary independently rather than within-city where they covary), do inequity features predict gun violence rates independently of race composition, or does race composition retain independent predictive power beyond inequity features?

Secondary question: Which inequity-feature effects generalize across geographic types (urban / suburban / rural) and which are specific to particular geographic mechanisms?

Methodological question: Can the structural-prior partial-pooling framework that identified BLK×Center coupling in Sloan and a_final residue classes in NBAProjections identify analogous structural couplings in gun violence rate data across heterogeneous geographic types?

§2. Cell definitions

Each cell is defined by demographics × inequity × geographic-type. Working-class American profile match (§3) applied within each cell.

Geographic granularity per cell type

- **Urban cells:** census tract (1,200–8,000 population). Population density homogeneity at tract scale is appropriate for urban areas.
- **Suburban cells:** census tract.
- **Rural cells:** county. Rural counties are functionally one community at the scale relevant to gun violence mechanisms.
- **Reservation cells:** reservation as primary unit, with multiple-tribe aggregation if needed.

Cell typology

Cell ID	Race composition	Inequity tier	Geographic type	Decoupling status
UB-HI	Black-majority (≥60%)	High	Urban tract	Coupled (literature standard)
UH-HI	Hispanic-majority (≥60%)	High	Urban tract	Coupled

UW-LI	White-majority (≥70%)	Low	Urban tract	Control
SB-MC	Black-majority (≥60%)	Low-to-moderate	Suburban tract	Decoupled: high race, low inequity
RW-HI	White-majority (≥80%)	High	Rural county	Decoupled: low race signal, high inequity
RB-HI	Black-majority (≥50%)	High	Rural county	Coupled (rural mechanism)
RH-HI	Hispanic-majority (≥50%)	High	Rural county	Coupled (rural mechanism)
RNA-HI	Native American (≥40%)	High	Reservation	Coupled (distinct historical mechanism)
Suburban transition	Mixed	Variable	Suburban tract	Reference for partial pooling

Cell inclusion criteria

Inequity tier is defined by composite z-score across four dimensions:

- Poverty rate (ACS B17001)
- Food access (USDA Food Access Research Atlas — low-income low-access status)
- School resource ratio (NCES per-pupil expenditure, student-teacher ratio at district level mapped to tract/county)
- Housing instability (ACS B25070 rent burden, B25003 owner/renter, eviction rate where available)

High inequity: composite z-score $\geq +1.0$ across at least 3 of 4 dimensions.

Low inequity: composite z-score ≤ -0.5 across at least 3 of 4 dimensions.

Moderate (excluded from primary analysis): composite z-score between -0.5 and +1.0.

Population threshold

Census tracts: $\geq 1,500$ population (excludes extreme small-cell noise).

Rural counties: $\geq 3,000$ population (same reasoning).

Reservations: $\geq 1,000$ population if treated as standalone unit; otherwise aggregated.

§3. Working-Class American matching profile

Cells matched on the following dimensions to ensure cross-cell comparability on baseline characteristics that aren't the inequity features being tested.

Matching dimensions

(1) **Household income.** Median household income between 30th and 65th percentile of national distribution. Excludes both extreme poverty (which has different dynamics) and middle-class-and-above (which is outside the working-class definition).

(2) **Education distribution.** Working-class education profile: $\geq 40\%$ high school graduate, $\leq 30\%$ bachelor's degree, balance of some-college / associate's. Filters out both highly-educated affluent areas and areas with extremely low educational attainment that may have different dynamics.

(3) **Labor force participation.** Employed civilian labor force $\geq 55\%$ of working-age population. Filters out areas with extreme unemployment or out-of-labor-force populations.

(4) **Household structure.** Married-couple-with-children + single-parent + multigenerational households combined $\geq 50\%$. Filters out extreme outliers (e.g., college-town tracts with mostly student households).

(5) **Age structure.** Working-age population (18-64) $\geq 55\%$. Filters out extreme outliers (retirement communities, military bases).

(6) **Stable population.** Population change between recent ACS 5-year estimates $\leq 25\%$ absolute. Filters out areas in rapid demographic transition that may have unstable outcome measurement.

Matching algorithm

Coarsened Exact Matching (CEM). Each dimension discretized into 3-5 bins, exact match required across all bins.

Rationale: CEM is interpretable, robust, doesn't require parametric assumptions about propensity score functional form. Standard in observational epidemiology for cross-

cohort matching.

Alternative considered: Entropy Balancing. Rejected because it makes the matching less transparent and the structural-prior framework already handles continuous covariate adjustment in the model itself.

Expected match rates

Pre-matching cell sizes (from §1 estimates):

- UB-HI: 500-1000 tracts
- UH-HI: 500-1000 tracts
- UW-LI: 1000+ tracts
- SB-MC: 300-500 tracts
- RW-HI: 150-250 counties
- RB-HI: 60-100 counties
- RH-HI: 40-60 counties
- RNA-HI: 20-30 reservations

After CEM matching to working-class American profile, expect 40-60% retention. Post-matching cell sizes:

- UB-HI: 200-500 tracts
- UH-HI: 200-500 tracts
- UW-LI: 400-800 tracts (subsampled to balance design)
- SB-MC: 120-300 tracts
- RW-HI: 60-150 counties
- RB-HI: 25-60 counties
- RH-HI: 16-36 counties
- RNA-HI: 8-18 reservations

Subsampling for balance

UW-LI control cell is over-populated. After matching, subsample to size of largest non-control cell to prevent control cell from dominating the partial-pooling structure.

§4. Outcome variable

Primary outcome

Gun violence rate per 100,000 population per year, averaged over a 5-year window (2019-2023 or most recent 5 years with complete data).

Includes:

- Homicide by firearm (CDC WISQARS underlying cause of death, ICD-10 X93-X95, U01.4)
- Suicide by firearm (CDC WISQARS, ICD-10 X72-X74, Y22-Y24)
- Non-fatal injury by firearm (GVA incident records geocoded to unit)

Excluded:

- Police-involved shootings (different mechanism, separate cell typology needed)
- Accidental shootings (different mechanism, sparse counts)
- Mass shootings as a category (rare events, statistically unstable at cell level)

Secondary outcomes

For sensitivity analyses:

- Homicide rate only (without suicide)
- Suicide rate only (different mechanism, included as separate analysis)
- Non-fatal injury rate only (GVA dependent)

Data source per outcome component

Component	Source	Granularity	Lag
Homicide rate	WISQARS	County	1-2 years
Suicide rate	WISQARS	County	1-2 years
Non-fatal injury	GVA	Lat/lon → unit	Current
Aggregated rate	Constructed	Unit	Pinned to WISQARS lag

Reporting-rate adjustment

Cells differ in reporting infrastructure quality. Adjustment factor per cell type:

- Urban with good local PD reporting: 1.0 (reference)
- Urban with moderate reporting: 1.05-1.15
- Suburban: 1.10-1.20
- Rural county with media coverage: 1.20-1.40
- Rural county without strong media coverage: 1.30-1.60
- Reservation: 1.40-1.80 (highly variable, additional uncertainty)

Adjustment factors derived from existing validation studies (JAMA Network Open 2023 multi-source comparison; Trace methodology validation).

Critical: adjustment factor uncertainty itself is propagated through the structural-prior framework. The model treats reporting rate as a latent variable per cell type with informed prior. We don't pretend the adjustment is exact.

§5. Independent variables (predictors)

Race composition

Census ACS B02001 (race alone), B03003 (Hispanic origin):

- Percent non-Hispanic white
- Percent non-Hispanic Black
- Percent Hispanic (any race)
- Percent non-Hispanic Asian
- Percent non-Hispanic Native American
- Percent non-Hispanic other / multi

Inequity features (separated, not composite, for analysis)

Food access: USDA Food Access Research Atlas

- Low-income low-access (LILA) status at 0.5 / 1 / 10 / 20 mile thresholds
- Tract-level percent of population with limited supermarket access

School resources: NCES Common Core of Data

- Per-pupil expenditure (district level, mapped to unit via boundary file)
- Student-teacher ratio
- Percent of teachers with ≥ 5 years experience
- Title I school proportion

Property tax base: State assessor data (where accessible)

- Median property value (ACS B25077)
- Total assessed value per resident
- Property tax revenue per resident

Housing instability: ACS + supplementary

- Median gross rent as percent of household income (B25070)
- Eviction rate (Eviction Lab data where available)
- Homeownership rate
- Housing cost burden ($>30\%$ income on housing)

Historical mechanism markers (used for H_HISTORICAL_MECHANISM):

- HOLC redlining grade (Mapping Inequality project) — urban only
- Sundown town status (Loewen database) — applicable both urban and rural
- Voting Rights Act preclearance status (pre-2013) — applicable to relevant Southern counties
- Reservation status — for tribal cells

Geographic-type indicator

Categorical: Urban tract / Suburban tract / Rural county / Reservation

Geographic-mechanism indicator

For urban cells: redlining grade, historical Black migration patterns

For rural cells: economic-collapse history (coal, manufacturing, agriculture), historical Reconstruction-era violence, sundown town status

For reservations: tribe, treaty status, BIA jurisdiction status

§6. Pre-registered hypotheses

Each hypothesis specifies what the cross-cell pattern would look like under different causal structures.

H_INEQUITY (the existing-literature consensus position)

Statement: Inequity features predict gun violence rates. Race composition predicts only through correlation with inequity features. After conditioning on the inequity feature set, race adds no significant independent predictive power.

Predicted pattern in decoupled cells:

- SB-MC (Black-majority, low inequity): low gun violence rates, comparable to UW-LI
- RW-HI (white-majority, high inequity): high gun violence rates, comparable to other high-inequity cells
- Cross-cell variation in rates tracks inequity composite, not race composition
- Race coefficient in joint model approximately zero after inequity adjustment

Falsification criterion: Race composition contributes $\geq X\%$ additional variance beyond inequity features in joint model. Specific X to be set after power analysis but before outcome data inspection — current target $X = 15\%$.

H_RACE_PURE

Statement: Race composition independently predicts gun violence rates beyond what inequity features explain. Some causal pathway operates through race-specific mechanisms not captured by measured inequity features.

Predicted pattern in decoupled cells:

- SB-MC: gun violence rates elevated relative to UW-LI despite comparable inequity
- RW-HI: gun violence rates not as elevated as urban Black-majority cells despite comparable inequity
- Race coefficient retains significance after inequity adjustment

Falsification criterion: Race composition contributes $< 5\%$ additional variance beyond inequity features. The hypothesis would be substantially weakened.

H_INTERACTION

Statement: Race $\times$ inequity interaction effects predict gun violence rates. Neither

marginal effect is sufficient; the joint co-occurrence is what matters.

Predicted pattern:

- Maximum gun violence rates in cells where high-inequity Black-majority intersect (UB-HI > sum of marginal effects)
- Off-diagonal cells (SB-MC, RW-HI) have intermediate rates
- Interaction term is significant in joint model

Falsification criterion: Race × inequity interaction explains <3% additional variance beyond main effects.

H_HISTORICAL_MECHANISM

Statement: Specific historical mechanisms (redlining, Reconstruction violence, economic collapse, treaty violations) explain geographic variation in gun violence rates beyond contemporary inequity features. The mechanism, not the current condition, is the causal driver.

Predicted pattern:

- Cells with same contemporary inequity profile but different historical mechanisms have different gun violence rates
- Urban Black-majority high-inequity (post-redlining mechanism) ≠ Rural Black-majority high-inequity (post-Reconstruction mechanism) ≠ Rural white-majority high-inequity (post-industrial collapse mechanism)
- Historical mechanism markers add significant predictive power in joint model

Falsification criterion: Historical mechanism markers add <5% additional variance beyond contemporary inequity features.

H_GEOGRAPHIC_MECHANISM

Statement: Specific inequity features have different effects in different geographic types. The food-access effect in urban tracts works through different intermediate pathways than the food-access effect in rural counties.

Predicted pattern:

- Same inequity feature has different effect sizes (and possibly different signs) across geographic types
- Geographic-type × inequity-feature interaction is significant

- Findings stratified by geographic type rather than averaged

Falsification criterion: Geographic-type × inequity-feature interactions consistently <2% additional variance beyond main effects.

Outcome possibilities

The hypotheses are not mutually exclusive. The actual finding could be:

- H_INEQUITY + H_GEOGRAPHIC_MECHANISM (inequity drives, but through different pathways by geography)
- H_HISTORICAL_MECHANISM + H_INTERACTION (historical mechanism produces specific contemporary inequity × race interactions)
- H_RACE_PURE + H_HISTORICAL_MECHANISM (residual race effect after inequity adjustment is explained by historical mechanism)
- Various other combinations

The pre-registration is what each hypothesis WOULD predict, not which hypothesis we expect to find.

§7. Power-stratified claim framework

Each cell has different statistical power based on post-matching sample size. Claims about each cell are pre-stratified by the power tier.

Tier 1: Strong claims (well-powered cells)

Cells: UB-HI, UH-HI, UW-LI, SB-MC, RW-HI

Sample sizes: post-matching ≥120 units per cell

Claim language permitted: "Effect estimated at X with 95% CI [Y, Z]. Effect generalizes within this cell type."

Effect sizes detectable: ≥10% relative variance contribution with 80% power

Tier 2: Moderate claims (adequately-powered cells)

Cells: RB-HI, RH-HI

Sample sizes: post-matching 25-100 units per cell

Claim language permitted: "Effect consistent with [direction]. Estimated effect size X

but cell-specific power limits precision; interpret with caveats. Findings hypothesized but not independently confirmed at cell level."

Effect sizes detectable: $\geq 20\%$ relative variance contribution with 80% power

Tier 3: Hypothesis-generating (under-powered cells)

Cells: RNA-HI

Sample sizes: post-matching 8-18 units

Claim language permitted: "Observed pattern consistent with [direction]. Cell-specific findings hypothesis-generating only. Larger study needed for independent confirmation. Pattern contributes to overall structural prior but not separately tested."

Effect sizes detectable: $\geq 40\%$ relative variance contribution with 80% power, or strong qualitative patterns

Cross-cell claims

Claims about cross-cell patterns (the decoupling hypothesis itself) use the joint sample across all cells. The full sample is well-powered for cross-cell effects even when individual cells are not. Cross-cell claim power is calculated separately.

§8. Analytic plan

Step 1: Data assembly

Construct unit-level dataset with:

- Outcome: 5-year average gun violence rate per 100,000
- Race composition: 6 race/ethnicity categories
- Inequity features: 4 dimensions $\times$ multiple measures within each
- Historical mechanism markers
- Geographic-type indicator
- Reporting-rate prior (informed by validation literature)
- Working-class American matching status (binary, post-CEM)

Step 2: Descriptive cell-level statistics

Pre-analysis report:

- Cell size after matching
- Cell-level mean rates and variances
- Cell-level inequity-feature distributions
- Cell-level race-composition distributions

This is reported BEFORE any inferential analysis to allow methodological critique of the sample before findings color the interpretation.

Step 3: Hierarchical Bayesian model

Structural-prior partial-pooling framework. Model specification:

```
gun_violence_rate[unit] ~ Negative-Binomial(mean[unit], dispersion[unit])

log(mean[unit]) =
  intercept
+ cell_type_effect[cell_type[unit]]
+ race_effect[race_composition[unit]]
+ inequity_effect[inequity_features[unit]]
+ race_inequity_interaction[race_composition[unit], inequity_features[unit]]
+ historical_mechanism_effect[historical_markers[unit]]
+ geographic_type_inequity_interaction[geo_type[unit], inequity_features[unit]]
+ reporting_rate_adjustment[geo_type[unit]]
+ log(population[unit]) // exposure offset
```

Priors:

- Cell-type effects: hierarchical normal with shrinkage to overall mean
- Race effects: hierarchical normal
- Inequity effects: weakly informative normal centered on direction from existing literature
- Interactions: tighter shrinkage (regularizing prior to avoid overfitting)
- Reporting-rate adjustment: informed by validation studies

Inference via Stan (HMC) or PyMC. Multiple chains, convergence diagnostics, posterior predictive checks.

Step 4: Pre-registered tests

For each hypothesis (H_INEQUITY, H_RACE_PURE, H_INTERACTION, H_HISTORICAL_MECHANISM, H_GEOGRAPHIC_MECHANISM):

- Compute the pre-registered test statistic (variance contribution, effect size, interaction significance)
- Compare against the pre-registered falsification criterion
- Report disposition (supported / not supported / inconclusive)

Step 5: Robustness checks

- Alternative matching specifications (Mahalanobis distance, propensity score)
- Alternative inequity feature operationalizations
- Sensitivity to reporting-rate adjustment uncertainty
- Leave-one-state-out cross-validation
- Alternative outcome operationalizations (homicide only, suicide only)

Step 6: Decoupled-cell focus analysis

Specifically examine SB-MC and RW-HI cells (the decoupled cells the design exists for):

- Predicted rates under H_INEQUITY (low for SB-MC, high for RW-HI)
- Predicted rates under H_RACE_PURE (elevated for SB-MC, moderate for RW-HI)
- Observed rates
- Bayes factor for H_INEQUITY vs H_RACE_PURE based on decoupled-cell patterns specifically

Step 7: Presentation stratified by cell tier

Tier 1 cells get strong-claim language. Tier 2 cells get moderate-claim language. Tier 3 cells get hypothesis-generating language. No averaging across tiers without explicit weighting by power.

§9. Threats to validity and mitigations

Threat 1: Reporting-rate adjustment mis-specification

If the reporting-rate adjustment factors are systematically wrong, cells with under-

counted incidents will appear to have lower true rates than they do.

Mitigation: (a) Propagate adjustment uncertainty through the model. (b) Run sensitivity analysis with alternative adjustment specifications. (c) Cross-validate with WISQARS mortality (which has uniform coverage) where available.

Threat 2: Working-class profile mis-specification

If the matching profile inadvertently excludes important variation, findings restrict to a more specific population than intended.

Mitigation: (a) Pre-register the profile definition explicitly. (b) Report sample composition before analysis. (c) Sensitivity analysis with alternative profile cutoffs.

Threat 3: Geographic granularity mismatch

Urban tracts and rural counties differ in size by orders of magnitude. Aggregating at different scales can introduce artifacts.

Mitigation: (a) Explicit geographic-type indicator in the model. (b) Multi-scale partial pooling. (c) Report effects per-100,000 standardized.

Threat 4: Inequity feature collinearity

The inequity features are correlated with each other and with race composition. Identifying independent effects is challenging.

Mitigation: (a) Report joint model and marginal models. (b) Use cells where features decouple (the SB-MC and RW-HI cells exist for this). (c) Sensitivity analysis with feature subsets.

Threat 5: Causal interpretation

The design is observational. Findings are associational. Causal claims require additional assumptions.

Mitigation: (a) Frame findings as predictive associations, not causal effects. (b) Discuss causal interpretation explicitly in limitations. (c) Note that decoupled-cell design provides stronger evidence for some causal pathways than within-city literature but doesn't establish causality.

Threat 6: Political reception

Findings will be politicized regardless of methodology.

Mitigation: (a) Pre-registration with public timestamp before analysis. (b) Report

findings whatever they show. (c) Methodologically rigorous framing makes selective use harder. (d) Cross-cutting findings (some support each side) reduce one-sided framing power.

§10. Estimated timeline

Assuming focused execution:

- **Weeks 1-4:** Data assembly. GVA scraping (or use of dxzys/Gun-Violence-Data dataset), WISQARS download, Census ACS pull, USDA Food Atlas, NCES extraction, HOLC mapping, redlining data, eviction data where available.
- **Weeks 5-8:** Cell construction and matching. CEM implementation, post-matching sample size verification, descriptive statistics.
- **Weeks 9-12:** Pre-registration finalization with public posting. Lock the design.
- **Weeks 13-20:** Model implementation (Stan/PyMC), prior calibration, convergence verification.
- **Weeks 21-24:** Primary analysis with pre-registered hypotheses. Robustness checks. Decoupled-cell focus analysis.
- **Weeks 25-28:** Writeup. Methods section emphasizing pre-registration and matching design.
- **Weeks 29-32:** Submission to target journal with pre-registration linked.

Total: ~8 months focused execution. Can be compressed if multiple workstreams parallelize. Can extend if data integration encounters obstacles.

Target journals

Primary: American Journal of Public Health (AJPH). Methodologically rigorous, audience includes public health researchers and policy makers.

Alternative: Social Science & Medicine (where Uzzi-Pollack-Porter Baltimore work was published — direct conversation with that literature).

Alternative: Health Affairs (policy-oriented audience).

Alternative: JAMA Network Open (methodologically rigorous, broad reach).

§11. Methodology corpus integration

This study serves as the fifth (or sixth) substrate for Paper 6 (methodology corpus): “noise is misclassified structure” framework demonstrated across mathematics (Collatz), sports (NBAProjections), options (a_final_sp500), cancer (CancerResearch), and now gun violence cross-geography.

The structural-prior partial-pooling approach used here is the same machinery as:

- BLK×Center coupling identification in Sloan paper
- a_final residue class partial pooling in NBAProjections
- WonkaResolve archetype coupling
- Options moneyness × tenor structural priors

The decoupled-cells design extension (sampling where features vary independently) is a methodological contribution specific to this substrate that may transfer back to other corpus members.

§12. Pre-registration commitments locked

The following are locked before data analysis begins:

1. Cell typology and thresholds (§2)
2. Working-class American matching profile and CEM algorithm (§3)
3. Outcome variable construction (§4)
4. Independent variables and operationalizations (§5)
5. Pre-registered hypotheses with falsification criteria (§6)
6. Power-stratified claim framework (§7)
7. Analytic plan and model specification (§8)
8. Threats to validity and mitigations (§9)

Any changes after data analysis begins are post-hoc and must be reported as such with rationale.

User additions